■■ DEFCON-D Security Experiment Report

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Attribute	Value
Experiment ID	12
Experiment Name	e2e_traffic_flood_test_3
Traffic Profile	HEAVY
Expected Duration	30 seconds
Virtual Clients count	80
Request Throttling delay	0.05s

■ Host Performance Metrics Summary

Peak CPU Usage	94.1%
Peak Memory Usage	97.1%
Max Traffic Throughput	0 requests/sec
Max Concurrent Sockets	1 connections
Average Processing Latency	0.00 ms

■ Security Alerts Logged

Severity	Alert Details	Attacker IP
RED	Low IP distribution entropy: 0.00 bits (Targeted flood indicator)	127.0.0.1
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■■ Mitigation Actions Deployed

Action Type	Target IP	Block Timer
BLOCK_IP	127.0.0.1	30s
BLOCK_IP	127.0.0.1	30s

■ Experiment Conclusion & Defensive Action Summary

Based on collected metrics, the platform has successfully simulated client connections. Under elevated loading (Medium/Heavy profiles), request frequency and latency spikes trigger the anomaly engines. In Automatic defense mode, the mitigation systems successfully logged rate-limit offenses and temporary blacklist overrides to SQLite database, blocking the targeted flows. This experiment confirms the effectiveness of multi-tiered rate limiting and signature alerts in DDoS mitigation.